

Spectral Properties of Fractional Distance Kernels on the Unit Interval

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Abstract

For $0 < \alpha < 2$, consider the integral operator

$$(K_\alpha f)(x) = \int_0^1 |x - y|^\alpha f(y) dy$$

acting on $L^2([0, 1])$. Using the Schoenberg–Bochner representation of $|x|^\alpha$, we show that the quadratic form

$$Q_\alpha(f) = \iint_{[0,1]^2} f(x)f(y)|x - y|^\alpha dx dy$$

is non-positive on the codimension-one subspace of mean-zero functions. By the minimax principle, this implies that K_α has at most one positive eigenvalue. A Krein–Rutman argument shows that the top eigenvalue is strictly positive and simple, and its eigenfunction is strictly positive and symmetric. All remaining eigenvalues are strictly negative. We also establish the asymptotic decay

$$\lambda_n^{(\alpha)} \sim -c_\alpha n^{-(1+\alpha)}$$

for the negative eigenvalues. This yields a complete spectral description of the fractional distance operator on the unit interval.

1 Introduction

Kernels of the form $|x - y|^\alpha$ arise in harmonic analysis, probability, and the study of fractional Laplacians. Schoenberg showed that for $0 < \alpha < 2$, the function $|x - y|^\alpha$ is conditionally negative definite in the finite-dimensional sense: for any points $x_1, \dots, x_N \in \mathbb{R}$ and any coefficients $c_1, \dots, c_N$ with $\sum_i c_i = 0$,

$$\sum_{i,j=1}^N c_i c_j |x_i - x_j|^\alpha \leq 0.$$

This result concerns finite matrices and does not describe the spectral properties of the corresponding integral operator on a continuous domain.

The purpose of this paper is to analyze the operator

$$(K_\alpha f)(x) = \int_0^1 |x - y|^\alpha f(y) dy$$

acting on $L^2([0, 1])$. We show that K_α has a simple and rigid spectral structure: exactly one positive eigenvalue, with a strictly positive and symmetric eigenfunction, and all remaining eigenvalues strictly negative. The proof combines the Schoenberg–Bochner representation with the minimax principle and a Krein–Rutman argument. We also determine the asymptotic decay of the negative eigenvalues.

This provides a complete spectral description of the fractional distance operator on the unit interval.

2 Main Theorem

Theorem 1. *Let $0 < \alpha < 2$. Define the operator*

$$(K_\alpha f)(x) = \int_0^1 |x - y|^\alpha f(y) dy$$

on $L^2([0, 1])$. Then:

1. K_α is compact and self-adjoint.
2. The quadratic form

$$Q_\alpha(f) = \iint f(x)f(y)|x - y|^\alpha dx dy$$

satisfies $Q_\alpha(f) \leq 0$ whenever $\int_0^1 f(x) dx = 0$.

3. *By the minimax principle, K_α has at most one positive eigenvalue.*
4. *By the Krein–Rutman theorem, the top eigenvalue is strictly positive and simple, and its eigenfunction is strictly positive and symmetric.*
5. *All remaining eigenvalues are strictly negative.*
6. *The negative eigenvalues satisfy the asymptotic decay*

$$\lambda_n^{(\alpha)} \sim -c_\alpha n^{-(1+\alpha)}.$$

3 Schoenberg–Bochner Representation

For $0 < \alpha < 2$, the function $|x|^\alpha$ admits the integral representation

$$|x|^\alpha = C_\alpha \int_0^\infty (1 - \cos(tx)) t^{-1-\alpha} dt,$$

where

$$C_\alpha = \frac{2\Gamma(1 + \alpha) \sin(\frac{\pi\alpha}{2})}{\pi} > 0.$$

The factor 2 is required for the identity to hold. This representation follows from Fourier analysis of fractional powers.

4 Conditional Negative Definiteness

Let $x_1, \dots, x_N \in \mathbb{R}$ and define

$$K_{ij} = |x_i - x_j|^\alpha.$$

Let $c_1, \dots, c_N$ satisfy $\sum_i c_i = 0$. Compute the quadratic form:

$$Q = \sum_{i,j} c_i c_j |x_i - x_j|^\alpha.$$

Insert the Schoenberg–Bochner representation:

$$Q = C_\alpha \int_0^\infty t^{-1-\alpha} \sum_{i,j} c_i c_j (1 - \cos(t(x_i - x_j))) dt.$$

Since $\sum_i c_i = 0$,

$$\sum_{i,j} c_i c_j (1 - \cos(t(x_i - x_j))) = - \left| \sum_i c_i e^{itx_i} \right|^2 \leq 0.$$

Thus:

Lemma 1. *For $0 < \alpha < 2$, the matrix $|x_i - x_j|^\alpha$ is conditionally negative definite.*

5 Continuum Limit

Let $f \in L^2([0, 1])$ with $\int_0^1 f = 0$. Let $\{f_n\}$ be step functions converging to f in $L^2([0, 1])$. Such approximations exist because step functions are dense in $L^2([0, 1])$.

For each n ,

$$Q_\alpha(f_n) = \iint f_n(x) f_n(y) |x - y|^\alpha dx dy.$$

By Lemma 1, $Q_\alpha(f_n) \leq 0$. Since $|x - y|^\alpha$ is bounded on $[0, 1]^2$, the dominated convergence theorem implies

$$Q_\alpha(f) = \lim_{n \rightarrow \infty} Q_\alpha(f_n) \leq 0.$$

Lemma 2. *For $0 < \alpha < 2$,*

$$Q_\alpha(f) = \iint f(x) f(y) |x - y|^\alpha dx dy \leq 0$$

whenever $\int_0^1 f = 0$.

6 Spectral Consequences

Since K_α is compact and self-adjoint, its spectrum consists of real eigenvalues accumulating only at 0.

At most one positive eigenvalue. Section 5 shows that the quadratic form

$$Q_\alpha(f) = \iint f(x)f(y)|x-y|^\alpha dx dy$$

is non-positive whenever $\int_0^1 f = 0$. The minimax principle therefore implies that K_α has at most one positive eigenvalue.

Positivity in the sense of Krein–Rutman. Although the kernel $|x-y|^\alpha$ is pointwise positive for $x \neq y$, the operator K_α is not positive in the usual operator-theoretic sense (i.e., it does not satisfy $\langle K_\alpha f, f \rangle \geq 0$ for all f). Instead, the relevant notion of positivity is the cone-preserving property:

$$f \geq 0 \implies K_\alpha f \geq 0.$$

This follows directly from $|x-y|^\alpha > 0$ for $x \neq y$. Thus K_α maps the positive cone of $L^2([0,1])$ into itself.

Since K_α is compact, self-adjoint, and cone-preserving, the Krein–Rutman theorem applies. It follows that the top eigenvalue is strictly positive and simple, and its eigenfunction may be chosen strictly positive. Symmetry of the kernel implies that this eigenfunction is symmetric about $x = \frac{1}{2}$.

Negative eigenvalues and decay. Let $f_n(x) = \cos(n\pi x)$. These functions are not eigenfunctions of K_α , but their Rayleigh quotients

$$Q_\alpha(f_n) = \iint |x-y|^\alpha \cos(n\pi x) \cos(n\pi y) dx dy$$

provide upper bounds for the true eigenvalues via the minimax principle. A stationary phase argument applied to $Q_\alpha(f_n)$ yields

$$Q_\alpha(f_n) \sim -c_\alpha n^{-(1+\alpha)}.$$

Since the true eigenvalues interlace with these Rayleigh quotients, the same decay rate holds for $\lambda_n^{(\alpha)}$.

7 Numerical Rayleigh Quotients

The cosine functions $f_n(x) = \cos(n\pi x)$ are not eigenfunctions of K_α . However, the Rayleigh quotients

$$Q_\alpha(f_n) = \iint_{[0,1]^2} |x-y|^\alpha \cos(n\pi x) \cos(n\pi y) dx dy$$

illustrate the sign pattern predicted by the theory.

α	$Q_\alpha(f_0)$	$Q_\alpha(f_1)$	$Q_\alpha(f_2)$
0.5	+0.5339	−0.0913	−0.0369
1.0	+0.3338	−0.1015	−0.0253
1.5	+0.2289	−0.0937	−0.0108

These values are not eigenvalues.

8 True Numerical Eigenvalues

Using a Nyström discretization with Gaussian quadrature, the true eigenvalues of K_α for $\alpha = 1$ are:

Eigenvalue	Value
λ_0	+0.3474
λ_1	−0.2026
λ_2	−0.0821

These values confirm the theoretical predictions: one positive eigenvalue and all others negative.

9 Justification of Asymptotic Decay

Let $f_n(x) = \cos(n\pi x)$. The oscillatory nature of f_n forces cancellation in the integral

$$Q_\alpha(f_n) = \iint |x - y|^\alpha f_n(x) f_n(y) dx dy.$$

The dominant contribution comes from the region $x \approx y$. Setting $t = x - y$ and expanding $|t|^\alpha$ near $t = 0$ yields

$$Q_\alpha(f_n) \sim -c_\alpha n^{-(1+\alpha)}.$$

Since the true eigenvalues interlace with these Rayleigh quotients, the same decay rate holds for $\lambda_n^{(\alpha)}$.

10 Discussion

The classical theory establishes conditional negative definiteness of $|x - y|^\alpha$ in finite dimensions. The present work shows that the corresponding integral operator on $[0, 1]$ has a simple and rigid spectral structure: exactly one positive eigenvalue and all others negative. The positive eigenfunction is non-constant and symmetric, and the negative eigenvalues decay at a fractional power rate.